3 The Hidden Shape

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> ***One-line motivation.*** *In Session 1, physics gave us the coordinates. What happens when we do not have a physicist in the room? If the coordinates mix the causes and we do not know the true basis, can we discover a useful representation purely from the data?*

Session 0 gave us the projectile. The failure there was that our *hypothesis class* could not express the shape of the phenomenon. Session 1 gave us the cantilever, where the *coordinates themselves* mixed the causes. In both cases, physics supplied the fix.

But what if we are handed a dataset from a system whose physics we do not understand? We only have a grid of numbers. We cannot write down the correct basis in advance. Do we have any hope of learning the representation from the data itself?

This is the question of Principal Component Analysis (PCA).

3.1 The Cantilever, Without Physics

Let us return to the cantilever from Session 1. We have two strain gauges at the same cross-section, measuring ε_{top} and $\varepsilon_{\text{bottom}}$. The physical causes are axial strain ε_a and bending strain ε_b.

$$\begin{bmatrix} \varepsilon_{\text{top}} \\ \varepsilon_{\text{bottom}} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}}_{\mathbf{A}} \begin{bmatrix} \varepsilon_a \\ \varepsilon_b \end{bmatrix} + \boldsymbol{\eta}, \tag{1}$$

where $\boldsymbol{\eta}$ is measurement noise.

In Session 1, we used the physics of beam deformation to invert $\mathbf{A}$ and find ε_a and ε_b. Now, let us pretend we never had that physics. We only see the raw readings $\mathbf{x}_{\text{raw}} = (\varepsilon_{\text{top}}, \varepsilon_{\text{bottom}})$.

3.2 The Meaning of Variance and Information

How do we find these axes? The key insight of PCA is to look for the direction in which the data varies the most.

Variance is a proxy for information. *If a feature has zero variance, it is a constant—it carries no information to distinguish one sample from another. If a feature has high variance, it changes drastically across samples, and thus is highly informative. PCA assumes that the most meaningful structure in the data lies along the directions of highest variance. By maximizing variance, we are preserving the maximum amount of information (in a least-squares sense) about the data.*

Suppose we want to project our n-dimensional data onto a single direction defined by a unit vector $\mathbf{u} \in \mathbb{R}^n$. Geometrically, we are dropping a perpendicular line from each data point down to the 1D line spanned by $\mathbf{u}$. The new coordinate of a data point $\mathbf{x}_i$ is simply its scalar projection $\mathbf{u}^T\mathbf{x}_i$.

If the data is spread out along the direction of $\mathbf{u}$, the projected points will have a large spread (variance). If the data is tightly clustered or spread perpendicular to $\mathbf{u}$, the projected points will collapse into a small cluster near the origin. **PCA seeks the specific direction u that maximizes the spread of these projected points.**

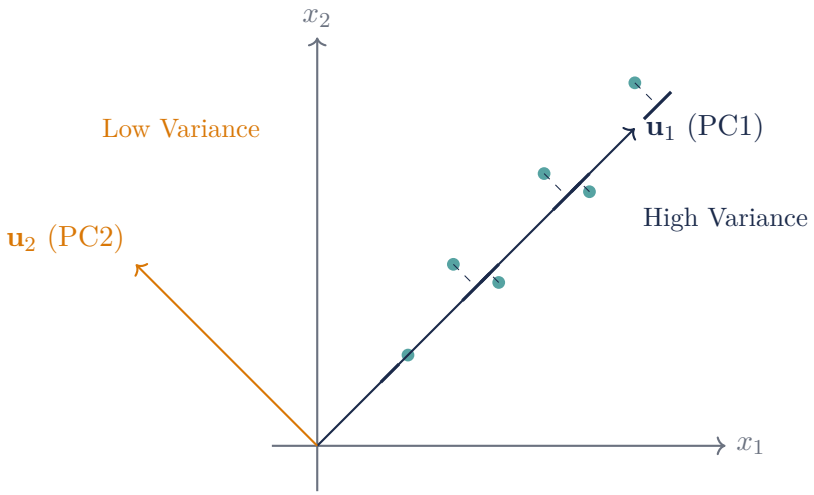


Figure 1: Projecting 2D data onto 1D. The variance of the projected points along $\mathbf{u}_1$ is much larger than along $\mathbf{u}_2$. PCA finds $\mathbf{u}_1$.

To find the best $\mathbf{u}$, we want to maximize the variance of this projection:

$$\text{Var}(\mathbf{u}^T\mathbf{X}) = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{u}^T\mathbf{x}_i)^2 = \mathbf{u}^T\left(\frac{1}{N}\sum_{i=1}^{N}\mathbf{x}_i\mathbf{x}_i^T\right)\mathbf{u} = \mathbf{u}^T\Sigma\mathbf{u}, \tag{2}$$

subject to the constraint that $\mathbf{u}$ is a unit vector: $\|\mathbf{u}\|^2 = \mathbf{u}^T\mathbf{u} = 1$. (Note that we assume the data $\mathbf{X}$ is already centered so the mean is zero).

3.3 The Covariance Matrix

Before we solve this optimization problem, let us pause and look at the matrix Σ, which is central to PCA.

Remark 3.1: The Covariance Matrix Σ

In PCA, the matrix Σ is the empirical covariance matrix of the centered data:

$$\Sigma = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i - \boldsymbol{\mu})^T$$

where $\boldsymbol{\mu}$ is the mean of the data.

Statistical Meaning:

- The **diagonal entries** Σ_{jj} represent the variance of the j-th feature. They tell us how much that individual feature varies.
- The **off-diagonal entries** Σ_{jk} represent the covariance between feature j and feature k. If $\Sigma_{jk} > 0$, the features tend to increase together. If $\Sigma_{jk} < 0$, one increases as the other decreases.

Geometric Meaning: The covariance matrix defines the shape of the data cloud. The diagonal entries define the width along the standard axes, while the off-diagonal entries define the tilt (rotation) of the ellipse. PCA seeks to diagonalize Σ, effectively removing the tilt and finding a coordinate system where the covariance between any two new axes is zero.

3.4 Deriving PCA

Now we are ready to derive the principal components. We want to maximize $\mathbf{u}^T\Sigma\mathbf{u}$ subject to $\mathbf{u}^T\mathbf{u} = 1$. To do that, we use the the properties of the covariance matrix. Since $\mathbf{u}^T\mathbf{u} = 1$ We want to maximize the Rayleigh quotient:

$$R(\mathbf{u}) = \frac{\mathbf{u}^T\Sigma\mathbf{u}}{\mathbf{u}^T\mathbf{u}}$$

Because Σ is symmetric and positive semi-definite, it can be diagonalized by its eigenvectors: $\Sigma = \mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}^T$, where $\mathbf{Q}$ is an orthogonal matrix whose columns are the eigenvectors $\mathbf{q}_1, \ldots, \mathbf{q}_n$, and $\boldsymbol{\Lambda}$ is a diagonal matrix of eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0$.

Any unit vector $\mathbf{u}$ can be expressed as a linear combination of these eigenvectors: $\mathbf{u} = \sum_{i=1}^{n} c_i\mathbf{q}_i$. Since $\|\mathbf{u}\|^2 = 1$, the coefficients must satisfy $\sum c_i^2 = 1$.

Substituting this into our variance equation:

$$\mathbf{u}^T\Sigma\mathbf{u} = \left(\sum_{i=1}^{n} c_i\mathbf{q}_i\right)^T \left(\sum_{j=1}^{n} \lambda_j\mathbf{q}_j\mathbf{q}_j^T\right) \left(\sum_{k=1}^{n} c_k\mathbf{q}_k\right) = \sum_{i=1}^{n} c_i^2\lambda_i$$

Because $\sum c_i^2 = 1$, this is a weighted average of the eigenvalues. To maximize this sum, we should put all our "weight" on the largest eigenvalue. That is, set $c_1 = 1$ and all other $c_i = 0$. Therefore, $\mathbf{u} = \mathbf{q}_1$, which is the eigenvector corresponding to the largest eigenvalue of Σ.

Proposition 3.1: PCA is an Eigenvalue Problem

The principal directions (components) of the data are the **eigenvectors** of the covariance matrix Σ. The variance of the data along each principal direction is given by the

corresponding **eigenvalue** λ.

Since Σ is symmetric and positive semi-definite, its eigenvectors are orthogonal, and its eigenvalues are real and non-negative. We sort the eigenvectors by their eigenvalues in descending order. The eigenvector with the largest eigenvalue λ_1 is the first principal component (PC1)—the direction of maximum variance. The second largest eigenvalue λ_2 corresponds to PC2, which is orthogonal to PC1 and captures the next highest variance, and so on.

To project the data onto a k-dimensional subspace, we simply form a matrix $\mathbf{W}$ whose columns are the top k eigenvectors of Σ. The new coordinates are $\mathbf{z} = \mathbf{W}^T\mathbf{x}$.

3.5 Evaluation Metrics: How Many Components Do We Need?

In the cantilever example, we had 2 dimensions, so we might just keep both. But in modern datasets with thousands of features, we must choose a smaller k. How do we evaluate how many components to keep?

Definition 3.1: Explained Variance Ratio (EVR)

The Explained Variance Ratio of the i-th principal component is the proportion of the total variance captured by that component:

$$\text{EVR}_i = \frac{\lambda_i}{\sum_{j=1}^{n} \lambda_j}.$$

The **cumulative EVR** is the sum of the top k ratios:

$$\text{Cumulative EVR}_k = \frac{\sum_{i=1}^{k} \lambda_i}{\sum_{j=1}^{n} \lambda_j}.$$

We typically plot a **scree plot** (a bar chart of EVRs) and choose k such that the cumulative EVR reaches a threshold, such as 0.95.

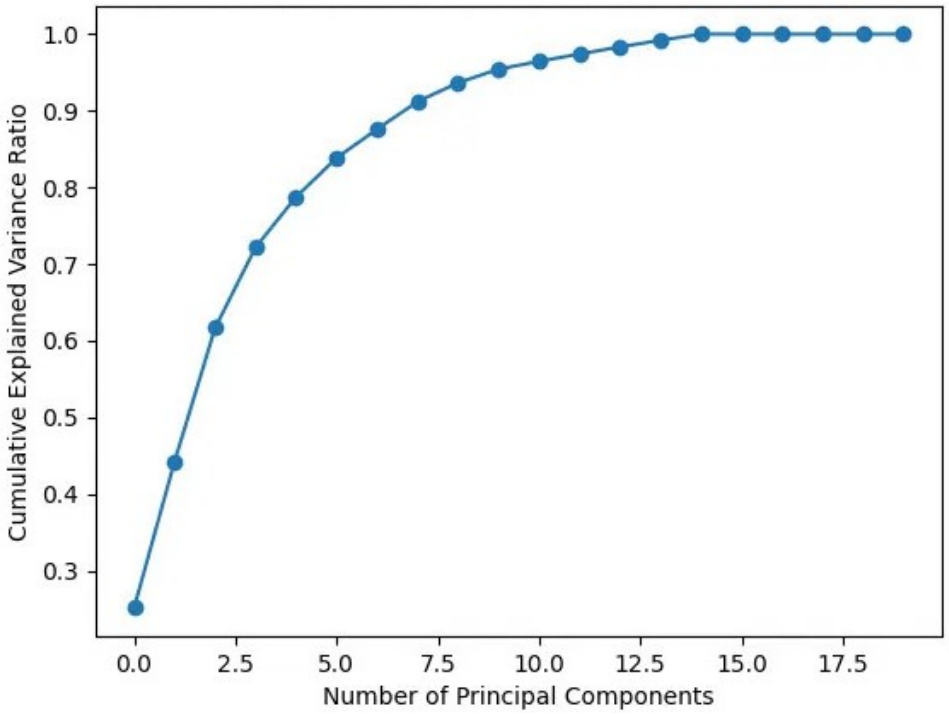


Figure 2: A Scree plot. The bars show the explained variance ratio for each component, and the line shows the cumulative variance. The "elbow" (where the curve flattens out) or a predefined threshold (e.g., 0.95) is typically used to select the number of components k to retain.

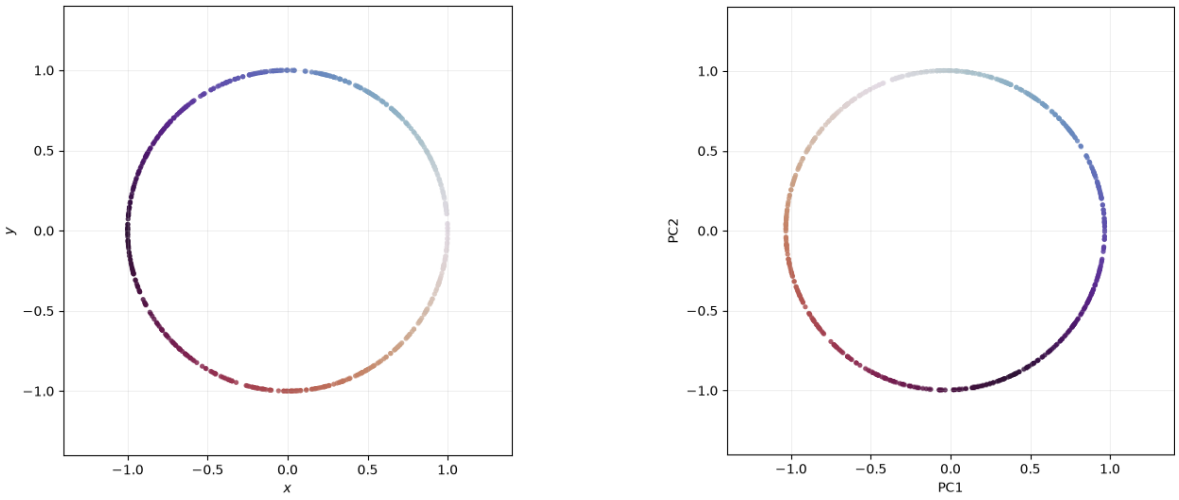
Another metric is **reconstruction error**. If we project data down to k dimensions and then rotate it back to the original space, how much information did we lose?

$$\text{Reconstruction Error} = \|\mathbf{x} - \mathbf{x}_{\text{reconstructed}}\|^2 .$$

The reconstruction of $\mathbf{x}$ using the top-k components is $\hat{\mathbf{x}} = \mathbf{W}\mathbf{W}^T\mathbf{x}$, and the reconstruction error is $\|\mathbf{x} - \hat{\mathbf{x}}\|^2$. PCA exactly minimizes this error for a given k.

Variance $\neq$ Meaning. In Session 1, we knew the physical axial axis because it was $\mathbf{u} = [1, 1]^T/\sqrt{2}$. If the axial variance is much larger than the bending variance, PCA will find the physical axis. But if the bending variance is larger, PCA will find the bending axis. If they are comparable, PCA will pick an arbitrary rotation. PCA is a variance engine, not a physics engine. It finds the shape of the data, not necessarily the shape of the phenomenon.

3.6 Failure Mode: Uncorrelated is Not Independent



(a) Raw Data (b) PCA Output

Figure 3: Left: Data points uniformly distributed on a circle $x = cos(\theta)$, $y = sin(\theta)$. The covariance between x and y is zero. Right: Because PCA only looks for linear covariance, it cannot find any preferred direction and simply retains the two components as equally important, failing to unfold the 1-dimensional manifold.

PCA finds directions that are *linearly uncorrelated*. It does this by diagonalizing the covariance matrix. But covariance only measures linear dependence. Two variables can be perfectly dependent and still have zero covariance.

Consider data lying on a circle: $x = \cos\theta, y = \sin\theta$. The covariance between x and y is exactly zero. PCA will see two equally important components and fail to find the 1-dimensional manifold (the circle itself).

> If the data lies on a non-linear manifold, PCA will completely miss it. The moment we move beyond linear relationships, PCA's assumptions break down.

3.7 Industrial Example: Sensor Fusion

In real-world systems, sensors are highly correlated. According to the **session's Jupyter notebook**, consider a wind turbine with 15 sensors (power, rotor speed, temperatures, vibrations) driven by 5 hidden physical drivers. If we apply PCA to standardized data, the cumulative EVR might show that 4 or 5 components capture 95 of the variance. By inspecting the *loadings* (the components of the eigenvectors), we can interpret what each PC means. For example, PC1 might heavily weight power and rotor speed (wind speed), while PC2 heavily weights vibrations (mechanical load).

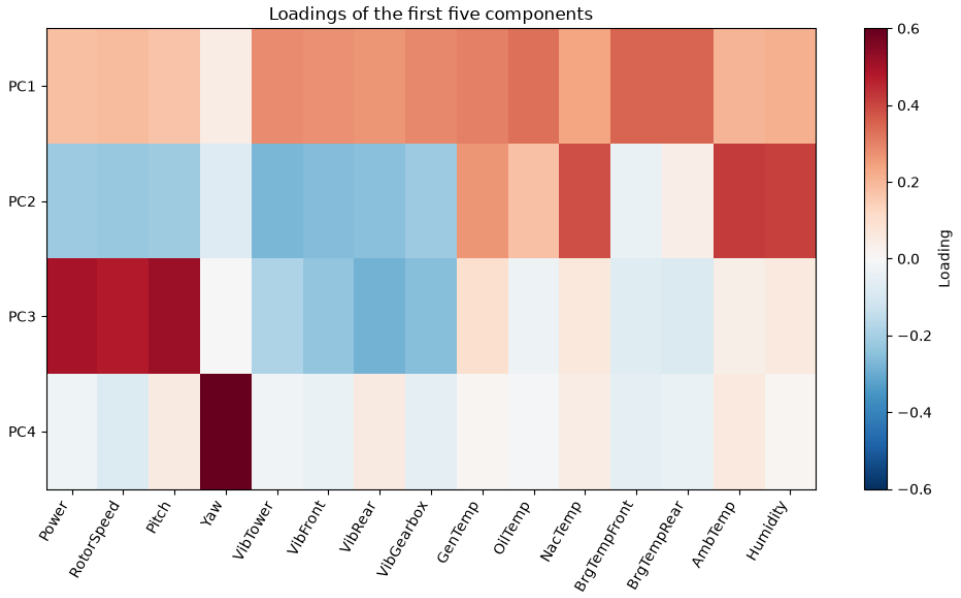
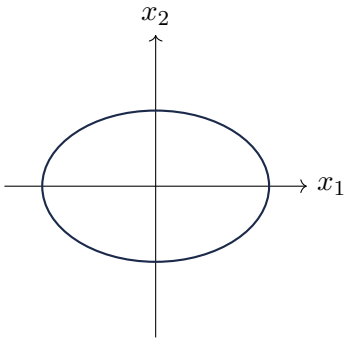


Figure 4: A heatmap of PCA loadings can reveal the hidden physical drivers in an industrial system.

3.8 Preprocessing: The First Step

PCA is extremely sensitive to the scale of the data. If one feature is measured in millimeters and another in kilometers, the variance of the former will dominate.

Therefore, we **always center** the data (subtract the mean) and usually **standardize** it (divide by the standard deviation).



Centered

3.9 What PCA Taught Us

PCA gives us a way to discover coordinates from data alone. It does so by finding the eigenvectors of the covariance matrix, which are the orthogonal axes of maximum variance. It is powerful when the data is linear and the important structure has high variance. But it fails when variance does not align with meaning, or when the data is non-linear.